EXPERIMENT-16

HISTOGRAM:

AIM:

To draw the histogram plot using R-tooll.

PROGRAM:

```
names<-c("siri","chru","loki")

age<-c(23,24,25)

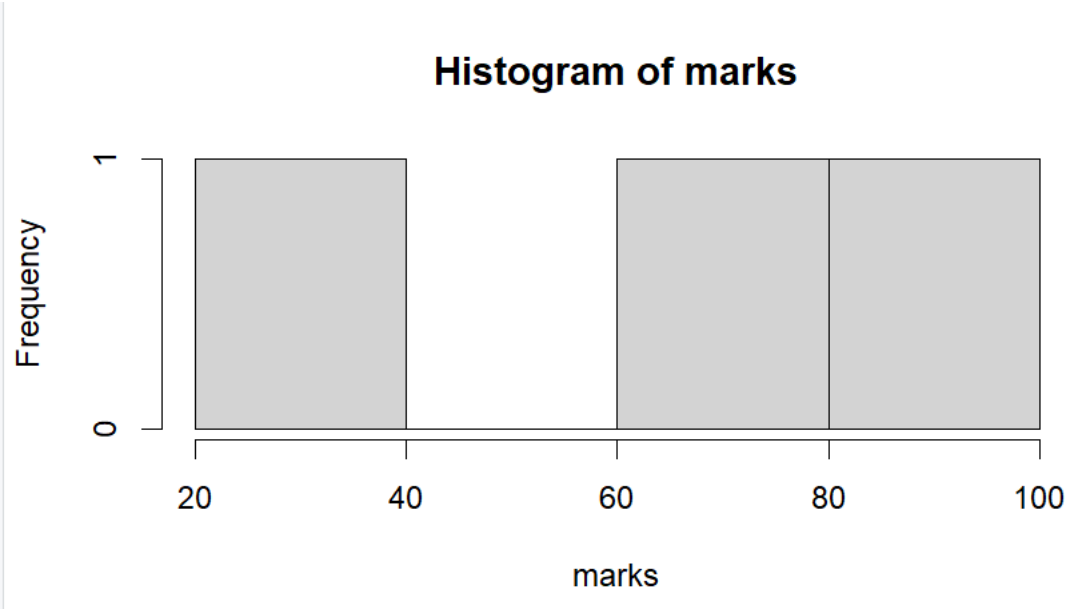
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

hist(age)
```

RESULT:

Thus the histogram plot was executed successfully



EXPERIMENT-17

CORRELATION ANALYSIS:

AIM:

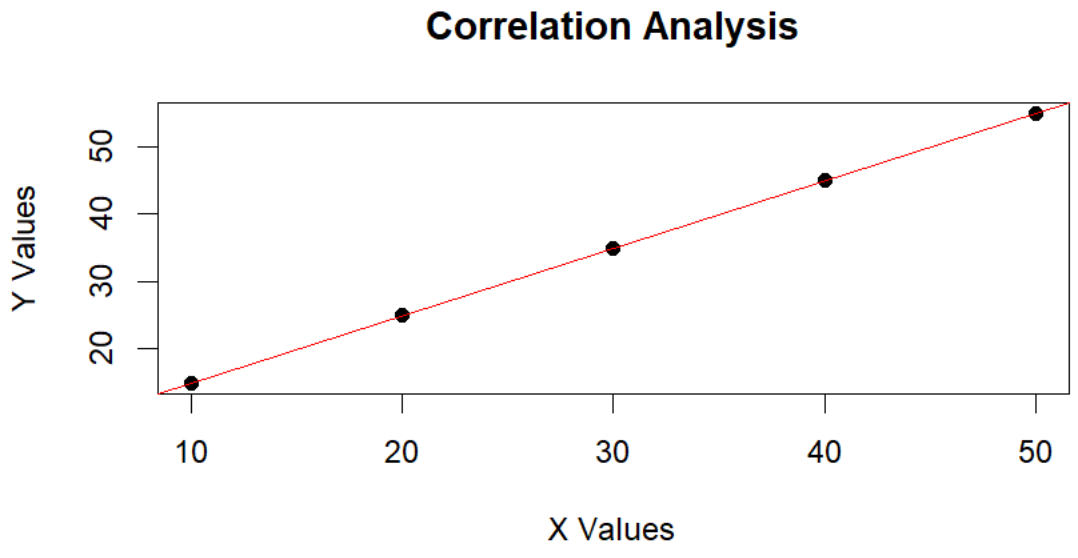
To write the program for correlation analysis using R-tool.

PROGRAM:

```
x <- c(10, 20, 30, 40, 50)
y <- c(15, 25, 35, 45, 55)
print(cor(x, y))
plot(x, y,
    main = "Correlation Analysis",
    xlab = "X Values",
    ylab = "Y Values",
    pch = 19)
abline(lm(y ~ x), col = "red")
```

RESULT:

Thus the correlation analysis was executed successfully.



SCATTER PLOT:

AIM:

To draw the scatter plot using R-tool

PROGRAM:

```
x <- c(10, 20, 30, 40, 50)

y <- c(15, 25, 20, 35, 45)

plot(x, y,

    main = "Scatter Plot",

    xlab = "X Values",

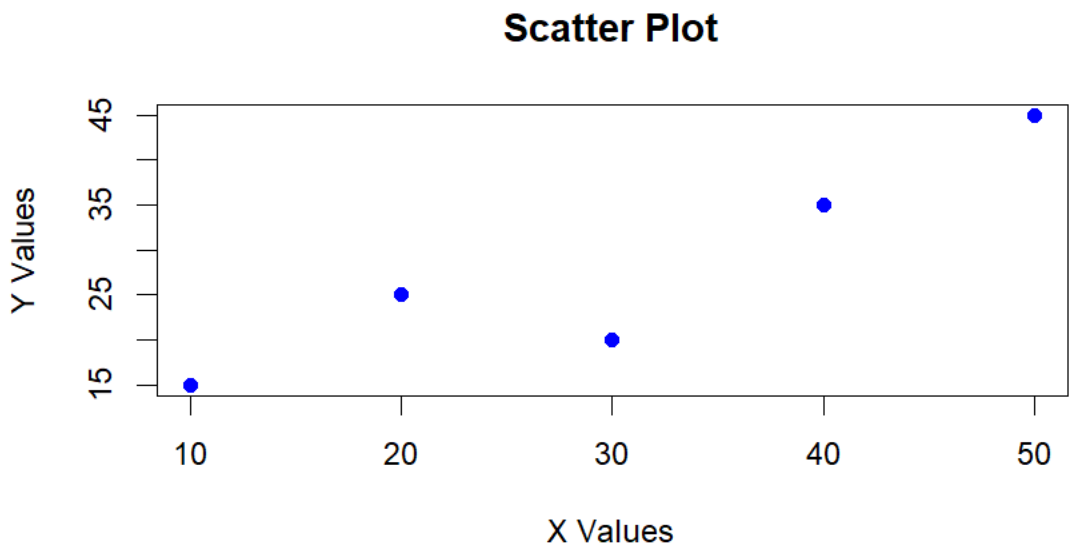
    ylab = "Y Values",

    pch = 19,

    col = "blue")
```

RESULT:

Thus the scatter plot was executed successfully.



EXPERIMENT-19

LINEAR REGRESSION:

AIM:

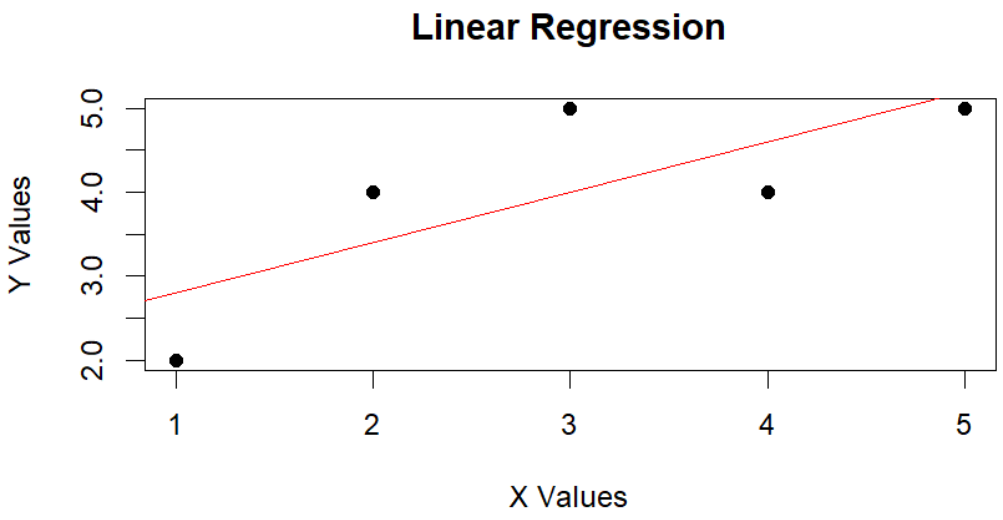
To write thr program for the linear regression using R-tool.

PROGRAM:

```
x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 5, 4, 5)
model <- lm(y ~ x)
print(model)
plot(x, y,
    main = "Linear Regression",
    xlab = "X Values",
    ylab = "Y Values",
    pch = 19)
abline(model, col = "red")
predict(model, data.frame(x = 6))
```

RESULT:

Thus the linear regression program was executed successfully.



EXPERIMENT-20

MULTIPLE REGRESSION:

AIM:

To write the program for the multiple regression.

PROGRAM:

```
area <- c(1000, 1500, 800, 1200, 2000)

bedrooms <- c(2, 3, 2, 3, 4)

price <- c(50, 75, 40, 60, 90)

data <- data.frame(area, bedrooms, price)

model <- lm(price ~ area + bedrooms, data = data)

print(model)

summary(model)

# Predict price for a new house

new_house <- data.frame(area = 1600, bedrooms = 3)

predict(model, newdata = new_house)
```

RESULT:

Thus, a multiple linear regression model was created successfully in R to predict house prices using area and number of bedrooms.

```
> area <- c(1000, 1500, 800, 1200, 2000)
> bedrooms <- c(2, 3, 2, 3, 4)
> price <- c(50, 75, 40, 60, 90)
>
> data <- data.frame(area, bedrooms, price)
>
> model <- lm(price ~ area + bedrooms, data = data)
>
> print(model)

Call:
lm(formula = price ~ area + bedrooms, data = data)

Coefficients:
(Intercept)         area     bedrooms
    8.50467      0.04252     -0.28037

> summary(model)

Call:
lm(formula = price ~ area + bedrooms, data = data)

Residuals:
      1       2       3       4       5
-0.4673  3.5514 -1.9626  1.3084 -2.4299

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.50467    6.54272   1.300   0.3233
area         0.04252    0.01261   3.373   0.0778 .
bedrooms    -0.28037    7.06829  -0.040   0.9720
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 3.486 on 2 degrees of freedom
Multiple R-squared:  0.9846,    Adjusted R-squared:  0.9692
F-statistic: 64.02 on 2 and 2 DF,  p-value: 0.01538

>
> # Predict price for a new house
> new_house <- data.frame(area = 1600, bedrooms = 3)
> predict(model, newdata = new_house)
       1
75.70093
```